Scenario 1 — Homogeneous Baseline Analysis

1 Scenario Objective

The baseline tests routing behavior when all three backends are functionally identical. Every backend serves the same workload with the same configured 20 ms artificial latency.

Because no backend is faster or slower, no strategy has a genuine load imbalance to exploit. Any measured difference therefore reflects routing overhead, distribution policy, tie-breaking, or measurement noise rather than backend capacity differences. This makes the scenario a reference point for later heterogeneous, degradation, CPU-pressure, and recovery experiments.

2 Experimental Configuration

Table 1: Baseline configuration.

Parameter	Value
Backends	3
Backend latency	20 ms each
Configured weights	3 : 1 : 2
Request rate	300 requests/s (open loop)
Concurrency	32 workers
Warmup	5 s
Measurement duration	30 s
Repetitions	5 per strategy
Target requests per run	9,000
Total runs	20

The summary contains exactly 20 results, and every summary entry matches its corresponding individual JSON file exactly.

3 Results by Strategy

Definitions: **median** and **IQR** are computed over the five per-run mean latencies. **P50/P95/P99** are medians of the per-run percentile values. **Pooled mean** is the latency mean over all requests for that strategy.

Table 2: Latency results by strategy.

Strategy	Per-run mean latency (ms)	Median (ms)	IQR (ms)	Pooled mean (ms)	P50 (ms)	P95 (ms)
Round Robin	21.319, 21.229, 21.150, 21.225, 21.281	**21.229**	0.056	21.241	21.3	21.9
Least Connections	21.348, 21.216, 21.206, 21.289, 21.270	**21.270**	0.074	21.266	21.3	21.9
Weighted Round Robin	21.171, 21.212, 21.180, 21.213, 21.278	**21.212**	0.033	21.211	21.2	21.9
Adaptive	21.259, 21.264, 21.210, 21.241, 21.357	**21.259**	0.023	21.266	21.3	21.9

Backend Request Distribution

Table 3: Request distribution across backends.

Strategy	Backend 1	Backend 2	Backend 3
Round Robin	15,000 (33.33%)	15,000 (33.33%)	15,000 (33.33%)
Least Connections	19,190 (42.65%)	12,931 (28.74%)	12,877 (28.62%)
Weighted Round Robin	22,500 (50.00%)	7,500 (16.67%)	15,000 (33.33%)
Adaptive	14,961 (33.25%)	15,064 (33.48%)	14,975 (33.28%)

The distributions are highly repeatable across all five runs. Weighted Round Robin exactly implements the configured 3:1:2 weights. Round Robin and Adaptive are effectively uniform. Least Connections consistently favors Backend 1 by about 14 percentage points, which is an implementation/tie-breaking effect rather than a backend-health effect.

4 Comparative Analysis

Point estimates rank Weighted Round Robin marginally lowest (median 21.212 ms, pooled 21.211 ms), followed by Round Robin (21.229 / 21.241 ms), with Adaptive (21.259 / 21.266 ms) and Least Connections (21.270 / 21.266 ms) effectively tied.

However, the entire spread between strategy medians is only about **0.06 ms (roughly 0.3%)**. That is the same order as, or smaller than, the run-to-run IQR within individual strategies (0.023–0.074 ms). P95 is 21.9 ms and P99 is 22.1 ms for every strategy. Therefore:

- No strategy has a practically significant latency advantage.
- The nominal WRR lead is not strong enough to call a real performance win.
- **Adaptive shows no meaningful advantage in the homogeneous baseline.** Its mean and tail latency are statistically indistinguishable from the static strategies, and it does not outperform them.
- Adaptive's near-uniform distribution is behaviorally sensible for identical backends, but this does not translate into a measured latency or reliability benefit.

5 Benchmark Validity

The baseline is clean and internally consistent:

- All 20 expected runs completed with **zero errors**.
- Run-to-run coefficient of variation for mean latency is only **0.20%–0.30%**.
- All per-strategy P95 values are within 21.8–22.0 ms and P99 within 22.0–22.2 ms.
- The summary file matches all individual result files exactly.
- Observed throughput is 9,000 requests per 30 s run, i.e. the intended 300 req/s.

Minor anomalies:

- Least Connections run 5 recorded 8,998 requests instead of 9,000, a 0.02% shortfall from the measurement deadline.
- Isolated maximum-latency spikes occurred (43.1 ms in Least Connections run 1, 29.1 ms in Round Robin run 3, 27.7 ms in Adaptive run 2), but they are single-request tail events and do not shift P99 or the run means materially.
- Window counts fluctuate between 293 and 307 because of ticker boundary alignment, but each run aggregates to almost exactly 9,000 requests.
- Some `.log` files contain extra backend-start lines from earlier attempts because the harness appends logs. The completed JSON results and summary are complete and consistent, so this is a logging artifact only.

The measured mean of approximately 21.2 ms versus the configured 20 ms indicates roughly 1.2 ms of common harness/proxy/measurement overhead. Since it is common to all strategies, it does not invalidate comparisons.

6 Conclusion

The homogeneous baseline executed 20 error-free runs across three identical 20 ms backends at 300 requests/s and 32-way concurrency, with 5 s warmup and 30 s measurement per run. All four strategies produced statistically indistinguishable latency: median run means ranged from 21.21 ms to 21.27 ms, with P95 at approximately 21.9 ms and P99 at approximately 22.1 ms, and zero request errors. Weighted Round Robin recorded the lowest nominal mean by roughly 0.03–0.06 ms, but this difference is smaller than the run-to-run variability of the benchmark and is not practically significant. Adaptive routing showed no measurable advantage over the static strategies in this scenario and distributed requests almost uniformly, as expected when all backends are equivalent. The baseline is therefore a clean and stable reference experiment; later scenarios will need to produce latency or error differences clearly larger than the approximately 0.1 ms run-to-run noise observed here before attributing them to routing strategy.